FoodVerse AI: Multimodal Food Intelligence & Precision Nutrition Agent

Template: Comprehensive Project Technical Documentation	**Maximum Marks: 5 Marks**
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Project Name: FoodVerse AI	**Track:** AI/ML & Generative AI

Executive Summary & Abstract

FoodVerse AI is an advanced full-stack multimodal food intelligence platform designed to eliminate the guesswork from nutritional tracking, reduce household food waste, audit commercial kitchen hygiene, and deliver safe, personalized clinical meal plans. Leveraging **Groq's Cloud AI infrastructure** with the **Qwen 3.8B Multimodal Vision-Language Model**, combined with real-time **OpenCV computer vision algorithms**, FoodVerse AI provides end-to-end intelligence across the entire food consumption lifecycle. The platform is implemented in Python FastAPI and React 18 / Vite 6, and is deployed globally on Vercel and Render.

Milestone 1: Model Selection & Technical Architecture

Activity 1.1: Groq API Key Setup & Credential Encapsulation
A secure Groq API key was generated and stored locally in backend/.env. The backend enforces strict credential protection by excluding .env from GitHub via .gitignore, while exposing safe template variables in backend/.env.example.

Activity 1.2: Research & Model Selection
Benchmark evaluations were conducted across available vision models. Groq's qwen/qwen3.8-27b was selected for multimodal plate recognition and video reasoning due to its sub-second latency, structured JSON reliability, and high visual acuity. Speech notes are transcribed using Groq's whisper-large-v3-turbo.

Activity 1.3: Define System Architecture
The architecture is organized into four distinct tiers: Presentation (React Vite on Vercel Edge), API Gateway (FastAPI on Render), Computational Vision (OpenCV headless), and Model Inference (Groq Multimodal Cloud).

Milestone 2: Development of Core AI & Vision Microservices

Activity 2.1: Multimodal Food Recognition & Macro Estimation
Implemented in routers/food_recognition.py. Accepts meal images and audio voice notes. Images are normalized to optimal resolutions (>=64px, <=1600px). Groq Qwen decomposes mixed plates, calculates calories, protein, carbs, fats, fiber, and allergens, and normalizes health scores to a 1-100 scale.

Activity 2.2: Computer Vision Spoilage & Browning Detection
Implemented in routers/freshness_detector.py. Applies Gaussian noise filtering, converts images to HSV color space, extracts brown/decay pixel masks, and calculates Laplacian gradient variance to measure surface roughness and cellular collapse.

Milestone 3: Commercial Kitchen Hygiene & Cooking Assistance

Activity 3.1: Video Frame Ingestion & Hygiene Auditing
Implemented in routers/hygiene_monitor.py. Ingests commercial kitchen video clips or photos, samples representative frames at calculated intervals, and inspects protective equipment (gloves, hairnets), prep surface sanitation, and cross-contamination hazards, assigning an audit grade (Grade A through F).

Activity 3.2: Interactive Culinary Assistant
Implemented in routers/kitchen_assistant.py. Generates structured, zero-waste recipes tailored to available pantry items and answers real-time cooking questions hands-free via voice or text.

Milestone 4: Precision Metabolic Meal Planning & User Interface

Activity 4.1: Metabolic Equations & Multi-Allergen Safeguards
Implemented in routers/meal_planner.py. Implements the Mifflin-St Jeor formula to determine Basal Metabolic Rate (BMR) and Total Daily Energy Expenditure (TDEE). Calibrates macronutrient targets based on health goals (weight loss, muscle gain, keto) and strictly excludes all declared allergens.

Activity 4.2: Frontend Workbench & Studio Tools
Built in React 18 with Vite 6. Features an interactive HTML5 canvas image generator (apple, banana, meal plate) for rapid headless testing and a Web Audio API microphone recorder.

Milestone 5: Verification, Testing & Public Cloud Deployment

Activity 5.1: Automated Edge-Case Validation
Validated with test_edge_cases.py, achieving a 100% pass rate (9/9 tests) across corrupted media, dim lighting, extreme ingredient scenarios, and multi-allergy constraints.

Activity 5.2: Public Cloud Deployment
Deployed to production with zero costs: Frontend hosted on Vercel at https://foodverse-ai-six.vercel.app/ and Backend hosted on Render with automatic GitHub CI/CD integration.